

improved system reliability. It also brought autonomy in optimal power storage and distribution."

As a result, the local administration was able to greatly reduce operational and vigilance cost for their streetlights. Solar power storage has been consistent and the setup has been running with minimal hiccups for nearly three years.

## What experts say

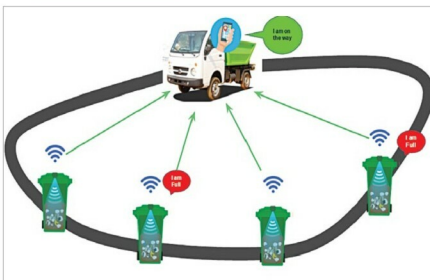
Bruno says, "The initial cost is a major factor for consumers in their choice of battery. The energy storage market for solar is still dominated by flooded tubular lead-acid batteries, as these are the cheapest among all the technologies. The selection is seldom based on the additional costs involved in maintenance. There is a small fraction of customers who understand the total cost of ownership and would opt for a lithium-ion or a gel battery, depending on their requirement. The other important points to consider are easy availability, warranty support and after-sales services."

—Paromik Chakraborty, technical journalist, EFM

# Smart Technology For Better Waste Management

▶ While Jharkhand's Jamshedpur city uses sensorised bins to track waste collection in real time, Gujarat's GIFT City has implemented an automated system for waste collection below ground

**S**ustainability and the level of living in a city can be judged through many factors. One of them is how city wastes are disposed



The concept of smart bins in smart waste management (Source: dqweek.com)

of or recycled. Specifically, for a densely populated country like India, waste generation is a major environmental concern. The country generates around 133.8 kilotonnes of solid waste every day, of which about 91 kilotonnes gets collected and only 25.9 kilotonnes is treated. While public awareness and education can bring down these figures, an effective waste management system can provide the much needed results.

Sensorised bins, automatic waste transportation, zero water discharge and many other solutions have been introduced for better waste management in India. Such waste management technologies help prevent piling up of excessive waste in an area. Sensors configured in waste collection bins offer real-time monitoring of the piled-up waste. This allows waste management authorities to send their clearance squad to the mentioned location and collect the pile without delay. As a result, overall hygiene and cleanliness of the area increases exponentially. Additionally, it reduces the time and effort required to travel to each bin and check the collection level manually.

Real-time navigation system for waste-collection vehicles ensures all collection bins in the specified area are collected—that too by covering the shortest route. This greatly cuts down fuel expenses and manual efforts. It can even allow municipali-

ties to cut down resources.

Many smart solutions allow automatic sorting of biodegradable and non-biodegradable waste. Automatic separation of waste like plastic, metal and glass can make the waste filtration process for recycling much easier. As a result, environmental damage due to non-biodegradable waste can be prevented. Manual sorting takes time and effort. More importantly, it is hazardous for human health and needs to be avoided.

Overall, smart waste management systems, if installed properly, can reduce financial burden for municipalities by up to 50 per cent. This includes resource requirement, transport cost, labour cost, activity monitoring and more. Additionally, smart waste management systems provide a streamlined accountability to contractual workers. Smart systems can allow municipal authorities to enroll them formally, keep better track of their whereabouts and record their activities.

The IoT in waste management is no longer a far cry. Smart waste systems are already in place throughout the globe and in India as well. For instance, the city of Jamshedpur has maintained zero liquid discharge from its sewage water wastes.

Ashish Mathur, MD, Jamshedpur Utilities and Services Company (JUSCO), explains, "Jamshedpur